didn’t, it would likely be moving at relativistic speeds in some direction that would take it far away from us in a very short time, because a collision is never so perfectly timed and aimed that the particles completely stop. Plus, for the kinds of collisions that happen in particle colliders to be capable of making tiny black holes, somehow the force of gravity felt by subatomic particles has to be stronger than Einstein’s laws of gravity would suggest. And the only way *that* could happen, as far as we know, is for there to be extra dimensions of space. We’ll talk more about this in the next chapter, but the short story is that having more than our usual three dimensions of space can make gravity a bit stronger on very small scales and can therefore allow LHC collisions to make little black holes.

So if we can make black holes in the LHC, we have evidence for space having more dimensions than we thought. Which to a physicist looking for signs of exciting new physics seems like it would be fantastic news! Of course, it would be a shame if those little black holes we are trying to make in the LHC could trigger vacuum decay and cause the universe to end …

Fortunately, they can’t. We are as close to absolute certainty about that as physicists ever get. The main thing that acquits them is the fact that, as previously stated, cosmic rays can make collisions much more powerful than anything we see in our own colliders. If we can smash protons together to make black holes, the universe has already done this countless times, and, look!—we’re still here! So either the black holes aren’t being made anywhere, or they were harmless all along.

The other reason is that it seems like there’s a mass threshold tiny black holes have to reach before they’re even hypothetically dangerous. The kinds of black holes a particle collider could make would be safely below that level, as would, most likely, many of the collisions that would happen out in space. As a side bonus, some of us have already been working on using this fact, along with our continued existence, as a way to argue that there have to be limits on the possible size extra dimensions could be.[fn9] (Just personally, as a cosmologist interested in testing different theories of physics, it’s always fun to be able to hold up the lack of a cosmic apocalypse as a data point.)

So, setting aside the little black holes for now, where does that leave us with vacuum decay? All the other possible ends of the universe we’ve looked at offer us at least the small comfort of being so far in the future that we can,